CUT_BEARING_CAPACITY PROGRAM THEORY AND USER MANUAL

Program: CUT_Bearing_capacity.py or CUT_Bearing_capacity.exe

Version: v7.1 (Program)

v3 (Manual)

Author: Dr Lysandros Pantelidis

Department of Civil Engineering and Geomatics

Cyprus University of Technology

About: Educational tool — no warranty. Use at your own risk. Free of charge.

CUT

BEARING CAPACITY

Table of Contents

1. Introduction 5
2. Computational Formulations 5
2.1. EN 1997-3 Drained Analysis (Effective Stress Analysis) 5
2.2. EN 1997-3 Undrained Analysis (Total Stress Analysis) 7
2.3. CUT Method 7
2.4. Mobilized Strength Analysis 11
2.5. Two-Layer Soil Profile 12
2.6. Punching and Local Shear Failure 14
3. Program Overview & GUI (graphical user interface) 16
3.1. Input Parameters Guide 16
Geometry & GWT 16
Soil Data 16
Loading Inclination 18
Ground Inclination 18
Seismic Excitation 19
Reinforced Earth (beta) 19
Resistance Factor Design 19
3.2. Running an Analysis 20
3.3. Outputs (right panel) 20
General overview 20
"Outputs" table organization 20
4. Generating a Report 23
5. References 23
6. Licensing, Copyright, and Disclaimer Statement 24
Licensing 24
NO WARRANTY 24
LIMITATION OF LIABILITY 25

1. Introduction

The CUT_Bearing_Capacity program is an educational software tool designed to calculate the ultimate bearing capacity of shallow foundations using two distinct methodologies:

(1) the EN 1997-3:202X (Eurocode 7), and

(2) the advanced CUT method.

The program applies to **uniform, homogeneous, and isotropic soils** consisting of a **single layer** with constant shear-strength parameters (c, ϕ, ψ) and unit weights ($\gamma, \gamma_{sat}, \gamma_w$). It accommodates **strip (plane-strain)**, **rectangular**, and **circular** footings, allowing for both **drained** and **undrained** conditions within the same computational framework.

2. Computational Formulations

2.1. EN 1997-3 Drained Analysis (Effective Stress Analysis)

General Bearing Capacity Equation:

$$q_{ult} = c' N_c b_c d_c g_c i_c s_c + q' N_q b_q d_q g_q i_q s_q + 0.5\gamma^* B' N_\gamma b_\gamma d_\gamma g_\gamma i_\gamma s_\gamma$$

Bearing Capacity Factors:

$$N_q = e^{\pi \tan\varphi'} \tan^2\left(45 + \frac{\varphi'}{2}\right)$$

$$N_c = (N_q - 1)\cot\varphi'$$

$$N_\gamma = 2(N_q + 1)\tan\varphi' \text{ for a rough base } \left(\text{i.e. } \delta \geq \frac{\varphi'}{2}\right)$$

Base Inclination Factors:

$$b_q = b_\gamma = (1 - \alpha \tan\varphi')^2$$

$$b_c = b_q - \left(\frac{1 - b_q}{N_c \tan\varphi'}\right)$$

Depth Factors:

$$d_\gamma = 1$$

$$d_q = 1 + 2\tan\varphi'\,(1 - \sin\varphi')^2 (D_f/B') \text{ for } \frac{D_f}{B'} \leq 1.0$$

$$d_q = 1 + 2\tan\varphi'\,(1 - \sin\varphi')^2 \tan^{-1}(D_f/B') \text{ for } D_f/B' > 1.0$$

$$d_c = d_q - \left(\frac{1 - d_q}{N_c \tan\varphi'}\right)$$

Ground Inclination Factors:

$$g_{\mathrm{q}} = g_{\gamma} = (1 - \tan\beta)^2$$

$$g_{\mathrm{c}} = g_{\mathrm{q}} - \left(\frac{1 - g_{\mathrm{q}}}{N_{\mathrm{c}} \tan\varphi'}\right) = \left(\frac{g_{\mathrm{q}} N_{\mathrm{q}} - 1}{N_{\mathrm{q}} - 1}\right)$$

Load Inclination Factors:

$$i_{\mathrm{q}} = \left(1 - \frac{T}{N}\right)^m$$

$$i_{\mathrm{c}} = i_{\mathrm{q}} - \left(\frac{1 - i_{\mathrm{q}}}{N_{\mathrm{c}} \tan\varphi'}\right) = \left(\frac{i_{\mathrm{q}} N_{\mathrm{q}} - 1}{N_{\mathrm{q}} - 1}\right)$$

$$i_{\gamma} = \left(1 - \frac{T}{N}\right)^{m+1}$$

$$m = m_B = \frac{2 + (B'/L')}{1 + (B'/L')} \text{ when } T \text{ acts in the direction of } B'$$

$$m = m_L = \frac{2 + (L'/B')}{1 + (L'/B')} \text{ when } T \text{ acts in the direction of } L'$$

$$m = m_\theta = m_L \cos^2\theta + m_B \sin^2\theta \text{ for other loading directions}$$

Shape Factors:

$$s_{\mathrm{q}} = 1 + \left(\frac{B'}{L'}\right) \sin\varphi' \text{ for a rectangular or circular } (B' = L') \text{ foundation}$$

$$s_{\mathrm{c}} = \left(\frac{s_{\mathrm{q}} N_{\mathrm{q}} - 1}{N_{\mathrm{q}} - 1}\right)$$

$$s_{\gamma} = 1 - 0.3\left(\frac{B'}{L'}\right) \text{ for a rectangular or circular } (B' = L') \text{ foundation}$$

Ground-Water Table Treatment:

- for groundwater level at ground surface:

$$q' = (\gamma_{sat} - \gamma_w) D_f \text{ and } \gamma* = \gamma_{sat} - \gamma_w$$

- for groundwater level at a depth D_w below the ground surface but above the foundation level:

$$q' = \gamma D_w + (\gamma_{sat} - \gamma_w)(D_f - D_w) \text{ and } \gamma* = \gamma_{sat} - \gamma_w$$

- for groundwater at the foundation level:

$$q' = \gamma D_f \text{ and } \gamma* = \gamma_{sat} - \gamma_w$$

- for groundwater at a depth exceeding 1,5 B below the foundation level:

$$q' = \gamma D_f \text{ and } \gamma* = \gamma$$

2.2. EN 1997-3 Undrained Analysis (Total Stress Analysis)

General Bearing Capacity Equation:

$$q_{ult} = c_{\mathrm{u}} N_{\mathrm{cu}} b_{\mathrm{cu}} d_{\mathrm{cu}} g_{\mathrm{cu}} i_{\mathrm{cu}} s_{\mathrm{cu}} + q_{\mathrm{o}}$$

Bearing Capacity Factors:

$$N_{\mathrm{cu}} = \pi + 2$$

$$(N_q = 1, N_\gamma = 0)$$

Inclination, depth, ground and shape factors:

$$b_{\mathrm{cu}} = 1 - \frac{2\alpha}{\pi + 2}$$

$$d_{\mathrm{cu}} = 1 + 0{,}33\tan^{-1}\left(\frac{D_{\mathrm{f}}}{B}\right)$$

$$g_{\mathrm{cu}} = \max\left(1 - \frac{2\beta}{\pi + 2}, 0\right)$$

$$i_{\mathrm{cu}} = \frac{1}{2}\left(1 + \sqrt{1 - \frac{T}{A' c_{\mathrm{u}}}}\right), T \leq A' c_{\mathrm{u}}$$

$$s_{\mathrm{cu}} = 1 + 0{,}2\left(\frac{B'}{L'}\right)$$

for a rectangular foundation or 1,2 for circular foundation

2.3. CUT Method

General Bearing Capacity Equation:

$$q_u = d_f \begin{pmatrix} cN_c(i_c s_c \varepsilon_{c,k} w_c b_c c_c) + \\ \frac{1}{2}\gamma B e_\gamma N_\gamma (i_\gamma s_\gamma \varepsilon_{\gamma,k} w_\gamma b_\gamma \mathrm{c}_\gamma) + \\ \gamma D_f N_q \left(i_q s_q \varepsilon_{q,k} w_q b_q c_q\right) + \\ q_r \end{pmatrix}$$

The CUT bearing capacity equation applies to both drained and undrained conditions. For drained analysis, $c = c'$ and $\varphi = \varphi'$; for undrained analysis, $c = c_u$ and $\varphi = \varphi_u = 0°$.

Virtual Wall Height and Depth of the Failure Mechanism:

Virtual Wall Height (unadjusted for embedment depth and dilation of the soil)

$$H = \frac{B}{2}\frac{e^{\left(\frac{\pi}{4} - \frac{\varphi}{2}\frac{\pi}{180}\right)\tan\varphi}}{\cos(45 + \varphi/2)}$$

Adjusted Virtual Wall Height

$$H^* = \kappa\left(1 + \lambda\frac{D_f}{B}\right)\frac{B}{2}\frac{e^{\left(\frac{\pi}{4}-\frac{\varphi}{2}\frac{\pi}{180}\right)\tan\varphi}}{\cos(45+\varphi/2)}$$

Max depth of the Failure Mechanism

$$H^*_{max} = \kappa\left(1 + \lambda\frac{D_f}{B}\right)\cos\varphi\,\frac{e^{\left(\frac{\pi}{4}+\frac{\varphi}{2}\frac{\pi}{180}\right)\tan\varphi}}{2\cos\left(45+\frac{\varphi}{2}\right)}B$$

Bearing Capacity Factors:

$$N_c = 2\frac{\sqrt{K_P}+\sqrt{K_A}}{K_A}\frac{H}{B}\frac{1}{J}$$

$$N_\gamma = \frac{K_P - K_A}{K_A}\left(\frac{H}{B}\right)^2\frac{1}{J}$$

$$N_q = \frac{D_{f,A}}{D_{f,P}} + \left(\frac{K_P}{K_A} - \frac{D_{f,A}}{D_{f,P}}\right)\frac{H}{B}\frac{1}{J}$$

$$N_r = \frac{1}{K_A}\frac{1}{J}$$

where,

$$J = \int_0^H I(z)dz\,/B = \begin{cases} -0.39\sin^3\varphi + 0.77\sin^2\varphi - 0.18\sin\varphi + 0.6 \text{ (rigid footings)} \\ 0.5\sin^2\varphi + 0.26\sin\varphi + 0.354 \text{ (flexible footings)} \end{cases}$$

Earth Pressure Coefficients (Coulomb's coefficients for smooth, non-battered wall):

$$K_A = \frac{cos^2\varphi}{\left[1 + \sqrt{\frac{sin\varphi \cdot sin(\varphi - \beta_A)}{cos\beta_A}}\right]^2}$$

$$K_P = \frac{cos^2\varphi}{\left[1 - \sqrt{\frac{sin\varphi \cdot sin(\varphi + \beta_P)}{cos\beta_P}}\right]^2}$$

For horizontal ground surface ($\beta_A = \beta_P = 0$), the above earth pressure coefficients are reduced to Rankine's expressions:

$$K_A = \frac{1 - sin\varphi}{1 + sin\varphi}$$

$$K_P = \frac{1 + sin\varphi}{1 - sin\varphi}$$

Shape Factors:

$$s_c = 1 + (3.16\tan^2\varphi + 0.27\tan\varphi + 0.16)\frac{B}{L} - (1.28\tan^2\varphi + 0.22\tan\varphi + 0.08)\left(\frac{B}{L}\right)^2$$

$$s_\gamma = 1 + (1.2\tan^2\varphi - 0.28\tan\varphi - 0.22)\frac{B}{L} - (0.2\tan^2\varphi + 0.16\tan\varphi - 0.06)\left(\frac{B}{L}\right)^2$$

$$s_q = 1 + 1.9\tan^2\varphi\sqrt{\frac{B}{L}}$$

Circular footing adjustment:

$$B = L = D$$

$$s_{c,circ} = \frac{4}{\pi}s_c$$

$$s_{q,circ} = \frac{4}{\pi}s_q$$

$$s_\gamma = \frac{4}{\pi}s_\gamma$$

For eccentric loading, a circular footing is converted to an equivalent rectangular footing with effective dimensions B' and L'. The problem is then solved as a rectangular foundation. Guidance for transforming the effective area of a circular foundation to an equivalent rectangular foundation is provided in EN 1997-3:202X.

Depth Factors:

$$\kappa = 1 \text{ for } \psi = 0$$

$$\kappa = 1.12 \text{ for } \psi = \varphi$$

$$d_c = d_q = d_\gamma = \min\left\{\kappa\left(1 + 0.2\frac{D_f}{B}\right), 1.4\right\}$$

Load Inclination Factors:

$$i_c = \left(1 - f_1 \cdot \tan\left(\frac{45° + \varphi°/2}{100°} \cdot \delta\right)\right)^3$$

$$i_\gamma = \alpha_\psi\,[1 - f_1 tan|\delta|]^3$$

$$i_q = \left[1 - f_1 tan\left(\frac{2\delta}{3}\right)\right]^3$$

$$\alpha_\psi = \begin{cases} 1, 0° \le \delta \le 45° \text{ and any } \psi \text{ value} \\ 1.46, -45° \le \delta \le 0° \text{ and } \psi = \varphi \\ 1.36, -45° \le \delta \le 0° \text{and } \psi = 0 \end{cases}$$

$$f_1 = \frac{B/L}{(1 + B/L)^2}\cos^2\theta + \left(1 - 0.75\frac{B}{L}\right)\sin^2\theta$$

Seismic Factors:

$$\varepsilon_{c,k} = \varepsilon_H$$
$$\varepsilon_{\gamma,k} = \varepsilon_H^2$$
$$\varepsilon_{q,k} = \varepsilon_H$$

where,

$$\varepsilon_H = \left(1 - \frac{k_h}{k_{h,lim}}\right)^{0.2}$$

$$k_{h,lim} = \tan\varphi + \frac{c}{(1 \pm k_v)\gamma_{kh} H_{max}}$$

$$\gamma_{kh} = (1 - f)\,\gamma + f\gamma_{sat}$$

$$f = \begin{cases} 0, D_w \geq D_f + H_{max} \\ \dfrac{D_f + H_{max} - D_w}{H_{max}}, D_f \leq D_w \leq D_f + H_{max} \\ 1, D_w \leq D_f \end{cases}$$

Note: $\varepsilon_x = \varepsilon_{x,k} \cdot \varepsilon_{x,i} = \varepsilon_{x,k} \cdot i_x$, where $\varepsilon_{x,k}$ and $\varepsilon_{x,i}$ are the reduction factors accounting for soil inertia and structure inertia, respectively, and i_x is the corresponding inclination factor for the given bearing capacity term. Here, x is c, γ or q, denoting the cohesion, soil weight, and lateral surcharge contributions, respectively.

Ground-Water Table Factors:

- Groundwater table above the foundation level

$$\begin{cases} w_c = 1 \\ w_\gamma = \dfrac{\gamma'}{\gamma} \\ w_q = \dfrac{\gamma'}{\gamma} + \left(1 - \dfrac{\gamma'}{\gamma}\right)\dfrac{D_w}{D_{fP}} \text{ for } 0 \leq D_w \leq D_{fP} \end{cases}$$

- Groundwater table below the foundation level

$$\begin{cases} w_c = 1 \\ w_\gamma = \frac{\gamma'}{\gamma} + \left(2 - \frac{D_w - D_{fP}}{H^*}\right)\left(1 - \frac{\gamma'}{\gamma}\right)\frac{D_w - D_{fP}}{H^*} \text{ for } D_{fP} \leq D_w \leq D_{fP} + H^* \\ w_\gamma = 1 \text{ for } D_w > D_{fP} + H^* \\ w_q = 1 \end{cases}$$

Base Inclination Factors:

At present, the CUT_bearing_capacity program adopts the EN base inclination factors; however, research in this area is still ongoing.

2.4. Mobilized Strength Analysis

In this analysis, the program performs an iterative (trial-and-error) procedure to determine the average mobilized shear strength of the soil that produces the same ultimate bearing capacity—evaluated with $k_h = k_v = 0$—as that predicted by the CUT method using static shear strength under $\{k_h, k_v\}$ pseudo-static conditions. The process operates as follows:

1. Target Value:
   The program first computes the ultimate bearing capacity using the CUT method (q_{ult}^{CUT}) with static shear strength under $\{k_h, k_v\}$ pseudo-static conditions. This value serves as the target bearing capacity for the mobilized analysis.
2. Assumed Strength Reduction:
   The soil strength parameters are temporarily reduced by a Strength Mobilization Factor (SMF):

$$c_m = SMF \cdot c$$

$$\varphi_m = arctan(SMF \cdot tan\varphi)$$

   where c_m and φ_m are the mobilized (reduced) cohesion and friction angle, respectively.

3. Iterative Adjustment:
   The program recalculates the CUT bearing capacity using these mobilized parameters, assuming static conditions ($k_h = k_v = 0$).
   The SMF is automatically adjusted through successive iterations until:

$$q_{ult}^{CUT}(c_m, \varphi_m, k_h = k_v = 0) = q_{ult}^{CUT}(c, \varphi, k_h, k_v)$$

   within a predefined numerical tolerance.
4. Resulting Mobilized Strength**:**
   The final mobilized values c_m, φ_m, and the corresponding factors ($N, b, d, g, i, s, w, \varepsilon$, etc.) are then displayed in the third column.

Interpretation:

- Under static conditions (no seismic excitation)
  The procedure converges to ($SMF = 1.0$), meaning that the full shear strength is mobilized; thus, $c_m = c$ and $\varphi_m = \varphi$.
  This confirms that the static bearing capacity is governed by the peak soil strength.
- Under seismic conditions ($k_h > 0$)
  The seismic ε-factors reduce the CUT bearing capacity.
  The mobilized analysis then converges to a $SMF < 1.0$, yielding smaller c_m and φ_m values.
  These represent the reduced average shear strength mobilized during earthquake failure, consistent with a partial mobilization of strength due to seismic shaking.

In summary, the Mobilized shear strength column provides an equivalent set of soil parameters that reproduce the same bearing capacity as the full CUT method, enabling users to interpret the average mobilized strength of soil at failure — static or seismic — in a physically meaningful way.

2.5. Two-Layer Soil Profile

For a soil profile consisting of two horizontal layers, the program does not solve the footing directly by a fully stratified formulation in the main CUT routine. Instead, it converts the two-layer system into an equivalent homogeneous soil and then applies the standard CUT bearing-capacity equations with the equivalent parameters c_{eq}, φ_{eq} and γ_{eq}. The upper layer has thickness H_1, whereas the lower layer is taken as extending to infinite depth.

The procedure implemented in the program follows the internal routine compute_two_layer_equivalent( ). The footing width used in the transformation is the effective width B, and the active- and passive-side embedment depths D_{fA} and D_{fP} are retained explicitly in the surcharge terms.

Case 1: Failure mechanism contained within Layer 1

The first check compares the thickness of the upper layer with the depth that the CUT mechanism would reach if the soil were entirely equal to Layer 1. That reference depth is:

$$H_{hom,1} = \left(\frac{B}{2}\right) \cdot \frac{exp\left[\left(\frac{\pi}{4} - \frac{\varphi_1}{2}\right) tan\varphi_1\right]}{cos\left(45° + \frac{\varphi_1}{2}\right)}$$

If $H_1 \geq H_{hom,1}$, the failure mechanism is fully developed inside the upper layer and no homogenization is needed. Therefore:

$$c_{eq} = c_1, \quad \varphi_{eq} = \varphi_1, \quad \gamma_{eq} = \gamma_1$$

In this case, the program classifies the regime as homogeneous_layer_1.

Case 2: Mechanism crossing the interface

If the upper layer is thinner than the reference depth above, the mechanism intersects the second layer. An equivalent friction angle is first obtained from the intersection geometry:

$$t_1 = tan\left(45° + \frac{\varphi_2}{2}\right), \quad t_2 = tan\left(45° + \frac{\varphi_1}{2}\right)$$

$$H_I = \left(\frac{B}{2}\right) t_1 + H_1 \left(1 - \frac{t_1}{t_2}\right)$$

$$\varphi_{eq} = 2\, arctan\left(\frac{2H_I}{B}\right) - \frac{\pi}{2}$$

Using this equivalent friction angle, the corresponding CUT mechanism depth becomes:

$$H = \left(\frac{B}{2}\right) \cdot \frac{exp\left[\left(\frac{\pi}{4} - \frac{\varphi_{eq}}{2}\right) tan\varphi_{eq}\right]}{cos\left(45° + \frac{\varphi_{eq}}{2}\right)}$$

If this recomputed depth happens to satisfy $H \leq H_1$, the program again falls back to the Layer-1 solution. Otherwise, the footing response is evaluated by combining contributions from both layers.

Rankine earth-pressure coefficients and influence integrals

For each layer $i = 1, 2$, the routine uses Rankine coefficients:

$$K_{Ai} = tan^2\left(45° - \frac{\varphi_i}{2}\right), \quad K_{Pi} = tan^2\left(45° + \frac{\varphi_i}{2}\right)$$

The weighting of each layer is based on the closed-form influence integral:

$$I(z_1, z_2) = F(z_2) - F(z_1)$$

$$F(z) = \frac{B\ ln(B^2 + z^2) + z\ arctan\left(\frac{B}{z}\right)}{\pi}$$

This function represents the analytical primitive of the stress influence kernel associated with a strip footing under plane-strain conditions. It enables the exact computation of the integrals:

$$I_1 = \int_0^{H_1} \frac{B}{\pi(B^2 + z^2)}\, dz = F(H_1) - F(0)$$

$$I_2 = \int_{H_1}^{H} \frac{B}{\pi(B^2 + z^2)}\, dz = F(H) - F(H_1)$$

The function $F(z)$ therefore provides an exact and computationally efficient means of evaluating the depth-weighted influence of soil layers without resorting to numerical integration. Within the two-layer formulation, these integrals are used to construct the total influence term I_0, which governs the relative contribution of each layer to the bearing capacity.

Physically, the logarithmic term reflects the spatial decay of stress with depth, while the arctangent term captures the geometric spread of the stress field beneath the footing. Together, they describe the cumulative stress influence within the soil mass in a form consistent with Boussinesq-type solutions.

Hence:

$$I_1 = I(0, H_1), \quad I_2 = I(H_1, H), \quad I_0 = K_{A1}\, I_1 + K_{A2}\, I_2$$

Two-layer bearing-capacity terms

The cohesion, unit weight and surcharge contributions are separated by layer. The program evaluates:

$$N_{c1} = 2\left(\sqrt{K_{A1}} + \sqrt{K_{P1}}\right)\frac{H_1}{I_0}$$

$$N_{\gamma 1} = (K_{P1} - K_{A1})\frac{H_1^2}{B\ I_0}$$

$$N_{q1} = \left(\frac{D_{fA}}{D_{fP}}\right)K_{A1}\frac{I_1}{I_0} + \left(\frac{K_{P1}}{K_{A1}} - \frac{D_{fA}}{D_{fP}}\right)K_{A1}\ \frac{H_1}{I_0}$$

$$N_{c2} = 2\left(\sqrt{K_{A2}} + \sqrt{K_{P2}}\right)\frac{(H - H_1)}{I_0}$$

$$N_{\gamma 2} = \left(\frac{\gamma_2}{\gamma_1}\right)(K_{P2} - K_{A2})\frac{(H^2 - H_1^2)}{B\ I_0}$$

$$N_{q2} = \left(\frac{D_{fA}}{D_{fP}}\right)K_{A2}\frac{I_2}{I_0} + \left(\frac{K_{P2}}{K_{A2}} - \frac{D_{fA}}{D_{fP}}\right)K_{A2}\frac{H - H_1}{I_0} + \left(1 - \frac{\gamma_2}{\gamma_1}\right)(K_{P2} - K_{A2})\frac{H_1(H - H_1)}{D_{fP}\ I_0}$$

The corresponding two-layer ultimate bearing capacity is:

$$q_u^{2-layer} = c_1 N_{c1} + c_2 N_{c2} + 0.5\, B\, \gamma_1\left(N_{\gamma 1} + N_{\gamma 2}\right) + \gamma_1 D_{fP}\left(N_{q1} + N_{q2}\right)$$

Determination of equivalent parameters

After the stratified ultimate capacity has been computed, the program defines an equivalent homogeneous soil for use in the ordinary CUT formulation. In the true two-layer regime, the equivalent unit weight is taken as the arithmetic average:

$$\gamma_{eq} = 0.5\,(\gamma_1 + \gamma_2)$$

The equivalent friction angle is the geometrically derived φ_{eq} above. Then the same mechanism depth H is re-evaluated with Rankine coefficients based on φ_{eq}, and equivalent homogeneous factors $N_{c,eq}$, $N_{\gamma,eq}$ and $N_{q,eq}$ are formed. The equivalent cohesion is finally back-calculated so that the equivalent homogeneous soil reproduces the same ultimate bearing capacity:

$$q_u^{2-layer} = c_{eq}N_{c,eq} + 0.5\,B\,\gamma_{eq}N_{\gamma,eq} + \gamma_{eq}D_{fP}N_{q,eq}$$

$$c_{eq} = \frac{q_u^{2-layer} - 0.5\,B\,\gamma_{eq}N_{\gamma,eq} - \gamma_{eq}D_{fP}N_{q,eq}}{N_{c,eq}}$$

Additional parameter averaging used by the program

The dilation angle used later in the CUT calculations is averaged with the same influence-integral weights:

$$\psi_{eq} = \frac{\psi_1 I_1 + \psi_2 I_2}{I_1 + I_2}$$

For the saturated unit weight, the program keeps γ_sat,1 when the regime remains inside Layer 1, and uses $0.5\,(\gamma_{sat1} + \gamma_{sat2})$ when the mechanism crosses the interface.

The above equivalent-soil procedure is currently tied to the CUT formulation implemented in the program. It provides a practical way to include two-layer effects while preserving the existing single-layer computational framework. The quality of the approximation depends on the assumption of two horizontal layers and on the ability of the equivalent parameters to represent the actual stratified failure mechanism.

2.6. Punching and Local Shear Failure

In the present formulation, punching shear and local shear failure are interpreted through the **spatial distribution of shear strength within the failure mechanism**. Rather than assigning a single strength throughout the soil mass, the method distinguishes between **peak** and **critical** shear strength and introduces appropriate weighted-average values in the different zones of the mechanism. The active and passive areas are therefore treated differently, and the transition from general shear to punching or local shear is expressed through the progressive replacement of peak strength by reduced or critical-state strength.

The **peak shear strength** corresponds to the conventional Mohr–Coulomb parameters c and φ, representing the maximum resistance mobilized by the soil. The **critical shear strength** corresponds to the state at large strains, where dilation is exhausted and the soil deforms at constant volume. In the present approach, the critical-state parameters are defined as:

- $c_{cr} = 0$,
- φ_{cr} derived from the reduction of $tan\varphi$ through the mobilization factor.

Thus, the transition from peak to critical strength is governed entirely by the reduction of friction, while cohesion vanishes in the critical state.

Punching Shear

For punching shear, the reduction in shear strength is **non-uniform and concentrated in the outer parts of the mechanism**.

- In **Zone I** (immediately beneath the footing), the soil retains a strong contribution from **peak strength**,
- In the adjacent active region, the strength is represented by an **average between peak and critical values**,
- In the **passive area**, the soil is assumed to mobilize **critical-state strength**.

This allocation reflects a mechanism in which the footing penetrates downward, while the surrounding soil provides limited lateral confinement. The passive resistance is therefore significantly reduced, and the load transfer becomes predominantly vertical.

Local Shear Failure

For local shear failure, the strength reduction is **more moderate and more uniformly distributed** within the mechanism.

- The **active area** remains associated with **peak strength**,
- The **passive area** is described by a **gradual transition from peak to critical strength**, implemented through weighted averaging over Zones II and III.

In this case, the passive resistance is only partially degraded, and the mechanism remains closer to general shear, although with reduced overall capacity. The soil does not reach the critical state abruptly, but progressively approaches it with increasing distance from the footing.

Interpretation

The distinction between punching and local shear failure is therefore governed by the **allocation of peak and critical shear strength within the active and passive zones**:

- **General shear** → peak strength dominates throughout
- **Local shear** → partial transition from peak to critical in the passive area
- **Punching shear** → critical strength dominates in the passive area, with peak retained only near the footing

This unified treatment allows a continuous and physically consistent transition between failure modes without the need for empirical correction factors.

3. Program Overview & GUI (graphical user interface)

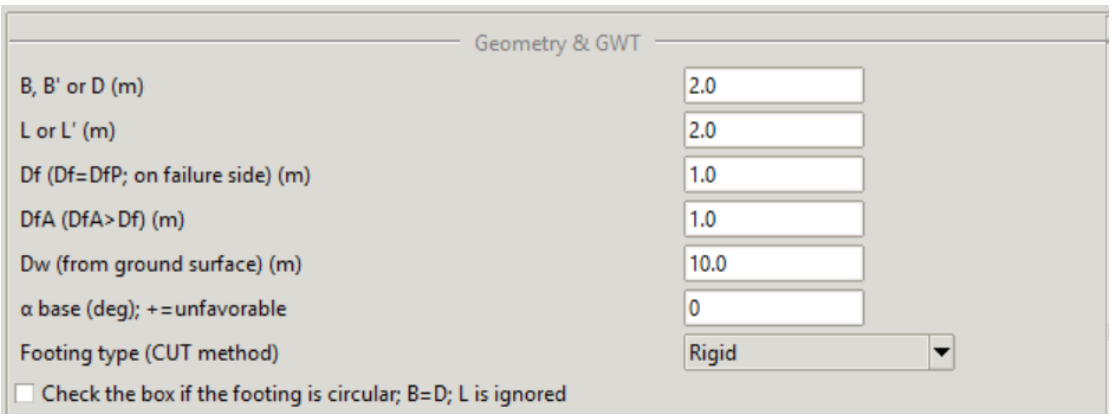
The GUI is split into two resizable panels:

(i) Left panel (Inputs) with grouped, scrollable fields;

(ii) (ii) Right panel (Outputs) showing tables of factors and q_{ult} for EN 1997-3, CUT method, and Mobilized Strength at pseudo-static excitation (CUT method).

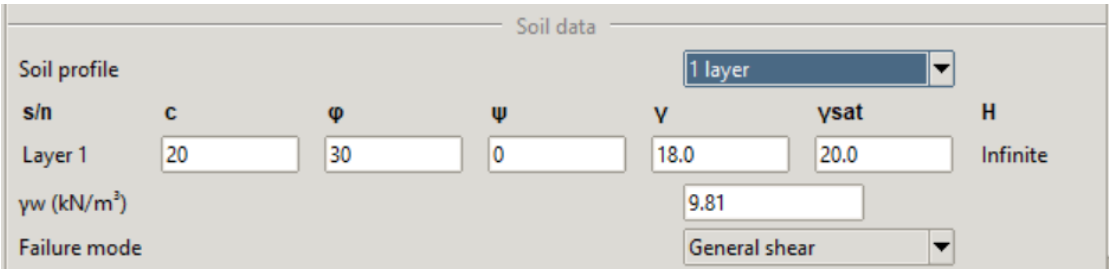
3.1. Input Parameters Guide

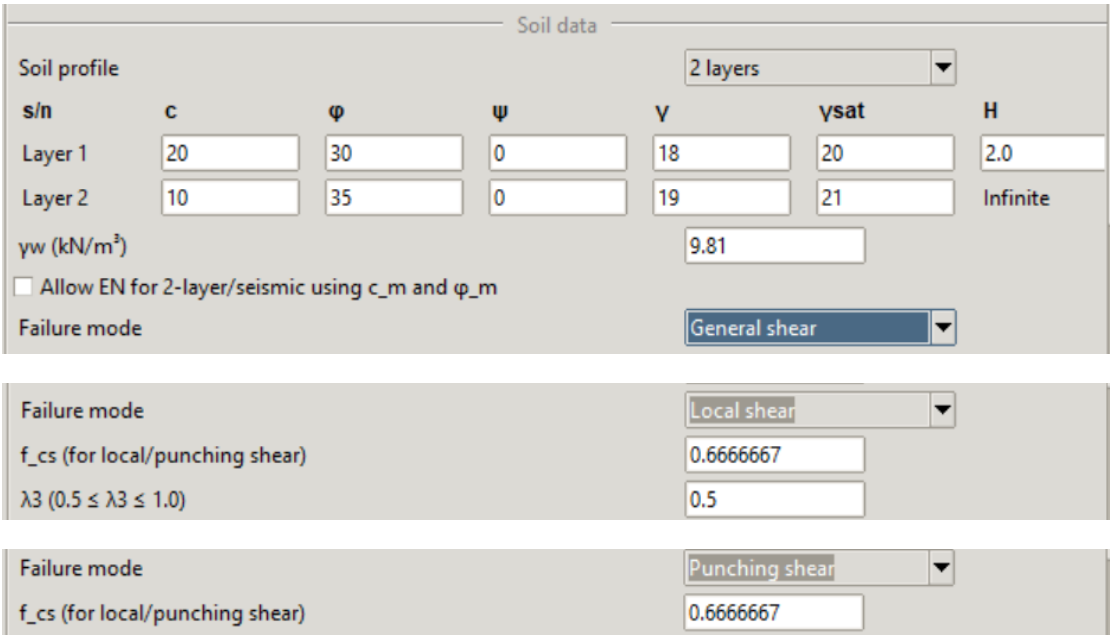
Geometry & GWT



- B, B' or D (m): Width, Effective width or Diameter of foundation.
- L or L' (m): Length or Effective length.
- D_f $(D_f = D_{fP};$ $on\ failure\ side)$ (m): Embedment at passive (failure) side.
- D_{fA} $(D_{fA} > D_f)$ (m): Embedment at active side.
- D_w (from ground surface) (m): Water table depth below ground surface.
- α base (deg): Base inclination.
- Footing type: Rigid or Flexible (CUT method). Choose from the drop-down menu.
- For circular footing tick the box. The footing be treated as circular with $B = D$; L is ignored.

Soil Data





Select **one-layer** or **two-layer** soil profile from the drop-down menu.

Insert the soil properties for each layer in the table provided:

- **c**: cohesion
- **φ**: friction angle
- **ψ**: dilation angle (deg)
- **γ**: bulk unit weight
- **γ_sat**: saturated unit weight
- **H**: thickness of the upper layer (Layer 2 extends to infinity)

For **drained conditions**, input the effective stress parameters $c = c'$, $\varphi = \varphi'$, and the corresponding dilation angle ψ.

For **undrained conditions**, input $c = c_u$, $\varphi = 0$, and $\psi = 0$.

The unit weight of water γ_w is defined separately.

When two layers are selected, the program accounts automatically for the variation of soil properties with depth.

The **failure mode** can be selected as:

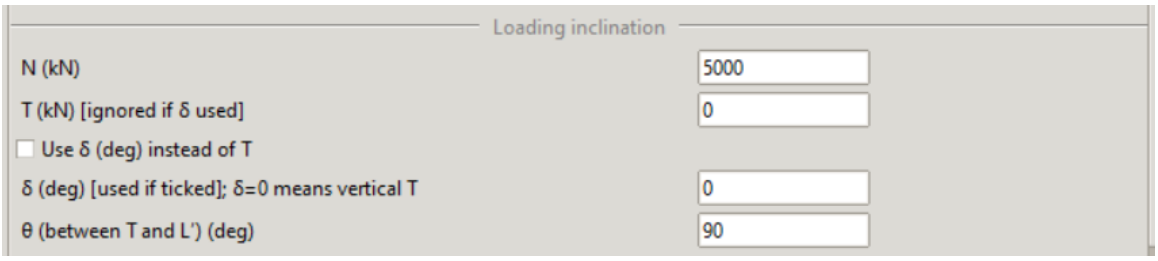
- **General shear**
- **Local shear**
- **Punching shear**

For **local** and **punching shear**, the reduction of shear strength is controlled through the parameter $\boldsymbol{f_{cs}}$ (strength mobilization factor).

For **local shear**, the parameter $\boldsymbol{\lambda_3}$ defines the extent of strength reduction in the passive zone.

The option **"Allow EN for 2-layer/seismic using c_m and ϕ_m"** enables the EN method to be applied in cases of layered soil or seismic conditions by using **mobilized (reduced) shear strength parameters** instead of the original input values.

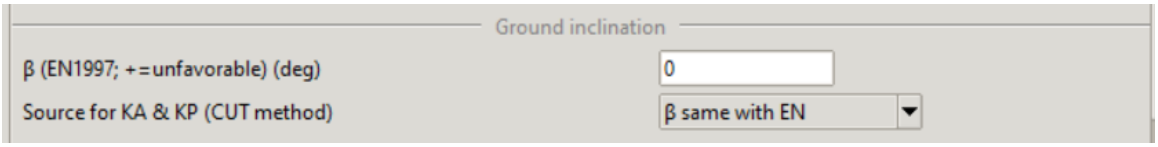
Loading Inclination



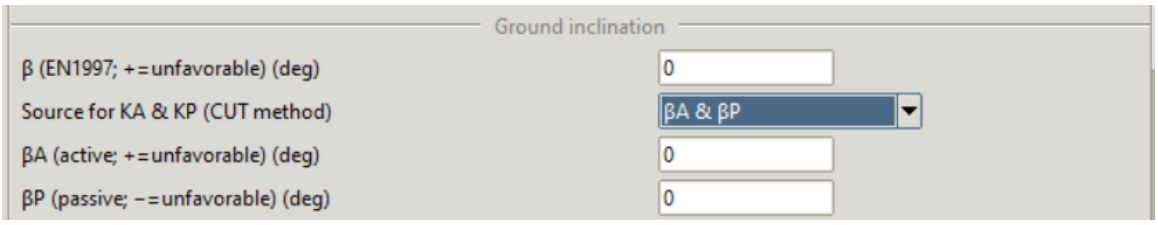
- N (kN): Vertical load
- T (kN): Horizontal load component.
- If the box is ticked, the program uses δ instead of T (ignored by the program) to define load by resultant angle.
- δ (deg): Inclination angle of resultant from vertical
- θ (deg): angle between T and L' direction.

Ground Inclination

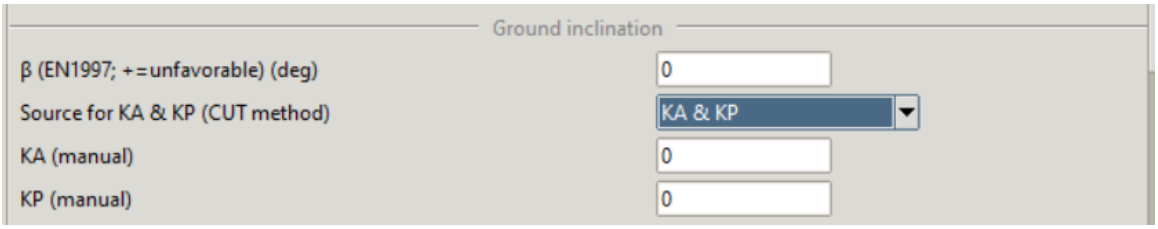
The EN method uses ground inclination factors, while the CUT method treats ground inclination with the Coulomb's earth pressure coefficients, K_A and K_P. The CUT method gives three options:



- β (EN1997) (deg): Ground slope for EN method.
- If the "β same with EN" is chosen from the dropdown menu, CUT method uses this value too.

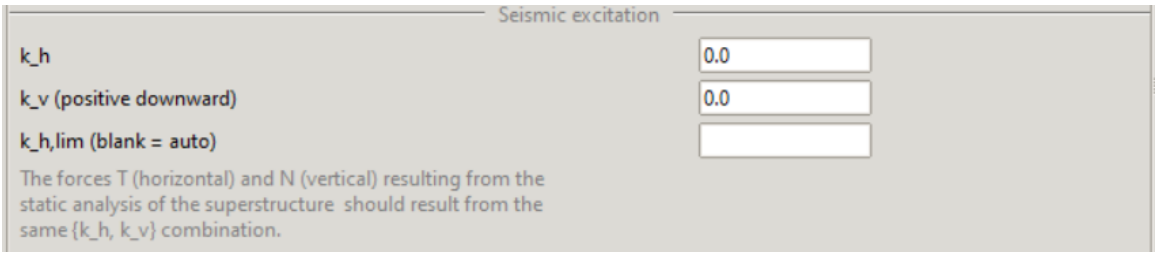


- If the "β_A & β_P" is chosen from the dropdown menu, CUT method uses different value for the active side (β_A) of the footing and different for the passive side (β_P) of the footing.
- If $\beta_A = \beta$ and $\beta_P = -\beta$ this option returns the same q_{ult} with the first option ("β same with EN")



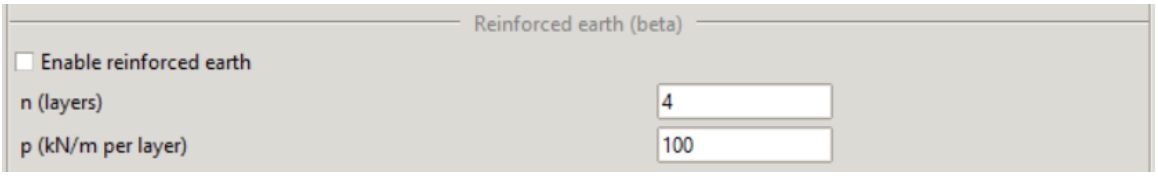
- If the "K_A & K_P" is chosen from the dropdown menu, the user can deal with any ground profile on both the active and passive sides. The auxiliary program **CUT_K_Coulomb** allows the user to draw these profiles point by point and calculate numerically the K_A and K_P values.

Seismic Excitation



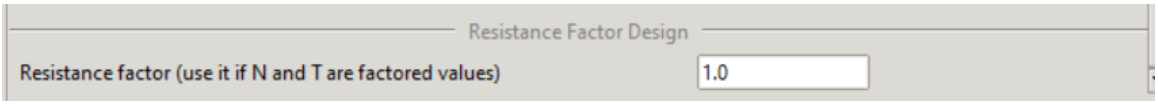
- k_h: Horizontal seismic coefficient
- k_v: vertical seismic coefficient (positive downward).
- $k_{h,lim}$: Limiting horizontal coefficient. If this field is left blank, it is calculated automatically by the program.
- The forces T (horizontal) and N (vertical) resulting from the static analysis of the superstructure should result from the same $\{k_h, k_v\}$ combination.

Reinforced Earth (beta)



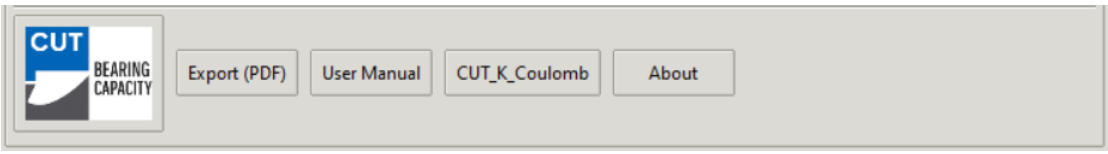
- Enable reinforced earth: activates reinforcement contribution.
- n (layers): number of layers; p (kN/m per layer): tensile capacity per unit width.

Resistance Factor Design



In a resistance factor design, q_{ult} is divided by the declared factor. The same factor is implemented to both EN and CUT methods.

3.2. Running an Analysis



After entering the input data, run calculations by clicking the Logo button.

3.3. Outputs (right panel)

General overview

In the Outputs panel of the program, the right-hand side of the interface displays a large, scrollable results table.

Each row corresponds to a specific factor or parameter used in the bearing-capacity calculation, and each column represents one of three parallel analyses:

Column	**Meaning**
1. EN 1997-3	Results calculated according to the Eurocode 7 (EN 1997-3:202x) bearing-capacity formulation. All factors—N, b, d, g, i, s —are evaluated following the EN method, with drained or undrained options as appropriate.
2. CUT method	Results from the Cyprus University of Technology (CUT) formulation. This column includes the CUT-specific factors such as modified *N*-factors, w-factors (for water-table effects), ε-factors (for seismic reduction), and any reinforcement contribution.
3. Mobilized shear strength of soil	Results of the mobilized-strength analysis based on the CUT method. Details are given below.

"Outputs" table organization

All intermediate parameters contributing to q_{ult} are displayed side-by-side for:

- the EN 1997-3 calculation,
- the CUT method (full-strength), and
- the CUT mobilized case, allowing a complete factor-by-factor comparison of how each approach affects the final bearing-capacity result.

The table is organized into groups separated by headings:

1. **— N_i factors —**
   Displays N_c, N_q, N_γ, and N_r *(reinforcement)*, the fundamental bearing-capacity

coefficients.
In the third (mobilized) column, these are recalculated using the mobilized c_m and φ_m values.

2. **— c_i (soil compressibility) —**
   Factors c_c, c_q, c_γ accounting for soil compressibility (punching shear or local shear failure; for general shear failure $c_c = c_q = c_\gamma = 1$).
3. **— b_i (base inclination) —**
   Factors b_c, b_q, b_γ accounting for foundation-base slope (α).
4. **— d_i (depth) —**
   Factors d_c, d_q, d_γ, and d_r, incorporating embedment effects (D_f/B) and dilation corrections.
5. **— g_i (ground inclination) —**
   Factors g_c, g_q, g_γ, representing the influence of ground slope (β).
   In the CUT method, slope effects are implicitly contained in the K_A/K_P coefficients, so g values appear as 1.0.
6. **— i (load inclination) —**
   Factors i_c, i_q, i_γ, accounting for load eccentricity and inclination (from T/N or δ).
7. **— s (shape) —**
   Factors s_c, s_q, s_γ, correcting for foundation shape (rectangular vs circular).
8. **— w (GWT) —**
   Factors w_c, w_q, w_γ, adjusting for groundwater-table position.
   (In EN these are shown as "incl. in γ*"; the CUT method computes them explicitly.)
9. **— ε (seismic) & $k_{h,lim}$—**
   Seismic reduction factors $\varepsilon_c, \varepsilon_q, \varepsilon_\gamma$, and the limiting horizontal acceleration $k_{h,lim}$ used to scale capacity under earthquake loading.
10. **— misc —**
    This group lists auxiliary parameters used internally by the calculations:

    - **δ used (deg):** the actual load-inclination applied (either the user-specified δ or derived from T/N).
    - **D_fA / D_fP**: ratio of embedment depths on the active and passive sides; influences the evaluation of the CUT bearing capacity factors.
    - **K_A used, K_P used**: active and passive earth-pressure coefficients adopted in the calculations.
    - **H*** (virtual wall height, depth-adjusted): effective height of the failure mechanism used in the CUT formulation.
    - **H*_max (depth-adjusted)**: maximum allowable virtual wall height after depth adjustment.
    - **Soil mode**: indicates whether the calculation is performed as a single-layer or equivalent two-layer problem.
    - **c_eq, ϕ_eq, γ_eq**: equivalent soil parameters used internally when a two-layer profile is converted into an equivalent homogeneous soil.
    - **Two-layer regime**: indicates whether the mechanism remains within the upper layer or extends into the second layer.
    - **EN special option**: shows whether the EN option using mobilized parameters $c_m$and $\varphi_m$is activated.
    - **EN status**: indicates whether the EN calculation follows the standard or modified formulation.

11. **— mobilized (SMF) —**
Displayed only in the third output column (Mobilized shear strength of soil).
It summarizes the mobilized-strength analysis based on the CUT method:

- $\boldsymbol{SMF}$ **(Strength Mobilization Factor):** automatically found so that the CUT bearing capacity computed with mobilized parameters equals the target CUT q_{ult} under static conditions ($k_h = k_v = 0$). Static → SMF ≈ 1.0; Seismic → SMF < 1.0.
- $\boldsymbol{c_m}$ **(kPa):** mobilized cohesion $= SMF \cdot c$.
- $\boldsymbol{\varphi_m}$ **(deg):** mobilized friction angle $= arctan(SMF \cdot \tan\varphi)$.

12. **— $\boldsymbol{q_{ult}}$, R_N, R_N/N —**
The final output block summarizes the overall bearing-capacity results for each analysis column:

- $\boldsymbol{q_{ult}}$ **(kPa):** ultimate bearing pressure, after applying the resistance factor to soil terms.
- R_N **(kN):** ultimate vertical bearing resistance = q_{ult}× Effective_Area (Effective Area $= B' \times L'$ or $\pi D^2/4$ for circular).
- R_N/N **(safety ratio):** ratio of ultimate resistance to applied vertical load; values >1.0 indicate sufficient safety under the specified conditions.

Outputs

Factor	EN 1997-3	CUT method	Mobilized shear strength of soil
— N factors —			
Nc	15.8679	28.6765	28.6765
Nq	7.1076	17.5564	17.5564
Nγ	6.2412	22.4002	22.4002
Nr (reinforcement)	n/a	4.5889	4.5889
— compressibility factors —			
cc	via strength	0.1455	0.1455
cq	via strength	0.3861	0.3861
cγ	via strength	0.3046	0.3046
Failure mode	Punching	Punching	Punching
— b (base inclination) —			
bc	1.0000	1.0000 (EN bc)	1.0000
bq	1.0000	1.0000 (EN bq)	1.0000
bγ	1.0000	1.0000 (EN bγ)	1.0000
— d (depth) —			
dc	1.1839	1.1000	1.1000
dq	1.1580	1.1000	1.1000
dγ	1.0000	1.1000	1.1000
dr	n/a	1.1000	1.1000
— g (ground inclination) —			
gc	1.0000	1.0000 (Nc; integrated)	1.0000
gq	1.0000	1.0000 (Nq; integrated)	1.0000
gγ	1.0000	1.0000 (Nγ; integrated)	1.0000
— i (load inclination) —			
ic	1.0000	1.0000	1.0000
iq	1.0000	1.0000	1.0000
iγ	1.0000	1.0000	1.0000
— s (shape) —			
sc	1.4180	1.7355	1.7355
sq	1.3592	1.6333	1.6333
sγ	0.7000	0.9193	0.9193

— w (GWT) —			
wc	incl. in γ*	1.0000	1.0000
wq	incl. in γ*	1.0000	1.0000
wγ	incl. in γ*	1.0000	1.0000
— ε (seismic) & k_h,lim —			
εc	n/a	1.0000	1.0000
εq	n/a	1.0000	1.0000
εγ	n/a	1.0000	1.0000
k_h,lim	n/a	0.8959	0.0000
— misc —			
δ used (deg)	0.0000	0.0000	0.0000
DfA/DfP	1.0000	1.0000	1.0000
KA used	n/a	0.3333	0.3333
KP used	n/a	3.0000	3.0000
H* (virtual wall height, depth-adjusted)	n/a	2.9765	2.9765
H* max (depth-adjusted)	n/a	3.4876	3.4876
Soil mode	1 layer	1 layer	1 layer
c_eq	20.0000	20.0000	20.0000
φ_eq	30.0000	30.0000	30.0000
γ_eq	18.0000	18.0000	18.0000
Two-layer regime	single_layer_input	single_layer_input	single_layer_input
EN special option	unticked	-	-
EN status	standard	-	-
— mobilized (SMF) —			
SMF	n/a	n/a	1.00000
c_m (kPa)	n/a	n/a	20.000
φ_m (deg)	n/a	n/a	30.000
q_ult (kPa)	**635.207**	**502.718**	**502.718**
R_N (kN)	**2540.827**	**2010.874**	**2010.874**
R_N / N (safety ratio)	**0.508**	**0.402**	**0.402**

4. Generating a Report

Click 'Export (PDF) to create a PDF containing inputs, factor tables, final bearing capacity values, program info, and timestamp.

5. References

Pantelidis, L. (2026). *Unified pseudo-static seismic reduction factors for shallow foundations via an earth-pressure framework.* Soils and Foundations, 66(1), 101713. https://doi.org/10.1016/j.sandf.2025.101713

Pantelidis, L. (2024). *Bearing capacity of shallow foundations: A focus on the depth factors in combination with the respective N-factors.* Arabian Journal of Geosciences, 17(6), 169. https://doi.org/10.1007/s12517-024-11976-7

Pantelidis, L., Assefa, E., & Sachpazis, C. I. (2025). *The effect of footing shape on the bearing capacity of shallow foundations: A review.* Archives of Computational Methods https://doi.org/10.1007/s11831-024-10184-6

Pantelidis, L. (2023). *Bearing capacity of centrically-vertically loaded strip foundations based on soil parameters: A classical earth pressure analysis problem.* Preprint (Version 2), Research Square. https://doi.org/10.21203/rs.3.rs-2940474/v2

Pantelidis, L., & Meddah, A. (2024). *The effect of loading inclination and eccentricity on the bearing capacity of shallow foundations: A review.* Archives of Computational Methods in Engineering, 31(7), 4189–4208. https://doi.org/10.1007/s11831-024-10113-7

Pantelidis, L., & Meddah, A. (2026). *Theoretical investigation of the effect of soil compressibility in the context of the bearing capacity of shallow foundations.* Proceedings of the 21st International Conference on Soil Mechanics and Geotechnical Engineering (ICSMGE 2026), Vienna, Austria, 14–19 June 2026.

Pantelidis, L., Assefa, E., & Sachpazis, C. (2026). *The effect of groundwater table on the bearing capacity of shallow foundations: An earth pressure analysis approach.* Proceedings of the 21st International Conference on Soil Mechanics and Geotechnical Engineering (ICSMGE 2026), Vienna, Austria, 14–19 June 2026.

6. Licensing, Copyright, and Disclaimer Statement

Licensing

This program is copyrighted © **2026 Dr. Lysandros Pantelidis**. It is distributed **free of charge for academic, educational, and research purposes only**.

Users may download, install, and employ the software **provided that proper citation of the original source and relevant publications is made**.

Modification, redistribution, commercial use, or incorporation into other software is strictly prohibited without the explicit written permission of the author.

All intellectual property rights relating to the program, its source code, algorithms, user interface, documentation, figures, and theoretical framework are fully retained by the copyright holder.

NO WARRANTY

THIS PROGRAM IS PROVIDED FREE OF CHARGE **"AS IS"**, WITHOUT ANY WARRANTY OF ANY KIND.

THE AUTHOR MAKES NO REPRESENTATIONS OR WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO:

- MERCHANTABILITY,
- FITNESS FOR A PARTICULAR PURPOSE,
- NON-INFRINGEMENT,
- ACCURACY OR RELIABILITY OF RESULTS.

THE ENTIRE RISK AS TO THE QUALITY AND PERFORMANCE OF THE PROGRAM IS WITH THE USER.

SHOULD THE PROGRAM PROVE DEFECTIVE, THE USER ASSUMES THE COST OF ALL NECESSARY SERVICING, REPAIR, OR CORRECTION.

LIMITATION OF LIABILITY

IN NO EVENT SHALL THE AUTHOR OR THE CYPRUS UNIVERSITY OF TECHNOLOGY BE LIABLE FOR ANY DAMAGES WHATSOEVER—INCLUDING, WITHOUT LIMITATION:

- GENERAL, SPECIAL, INCIDENTAL, OR CONSEQUENTIAL DAMAGES,
- LOSS OF DATA,
- LOSS OF PROFITS,
- BUSINESS INTERRUPTION,
- OR ANY CLAIM BY A THIRD PARTY—

ARISING FROM THE USE OR INABILITY TO USE THE PROGRAM, EVEN IF THE AUTHOR HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.